DVA PROJECT

FIFA World Cup Analytics Dashboard

Transforming 84 Years of Match Data into Competitive Intelligence

SECTOR

Data Analytics & Visualization

FACULTY MENTORS

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TEAM ID

Section D – Group 11

TEAM MEMBERS

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DATASET SCOPE

TOTAL GOALS

1,056

Outfield scorers · 1930–2014

YEARS OF DATA

84

1930 to 2014 World Cups

DASHBOARDS BUILT

3

Player · Match · Team-Country

Powered by **Tableau**

THE CHALLENGE

The Intuition Gap in Modern Football Strategy

Fragmented Data

Match, player, and team data siloed across disconnected systems — no unified analytical view.

IMPACT

3 Disconnected Sources

Delayed Insights

Patterns discovered only after matches conclude — reactive strategy instead of proactive intelligence.

COST

Post-Match Regret

No Real-Time Tactics

Raw stats never converted into live decisions — coaches rely on intuition over evidence.

OPPORTUNITY

Data-Driven Edge

DATA FOUNDATION

1930–2014 World Cup Records

TOTAL GOALS

1,056

Outfield scorers dominant

UNIQUE CAPTAINS

158

Across all tournaments

RECORD APPEARANCES

33

Miroslav Klose

FOREIGN COACH %

10.31%

International managers

3 TABLEAU DASHBOARDS

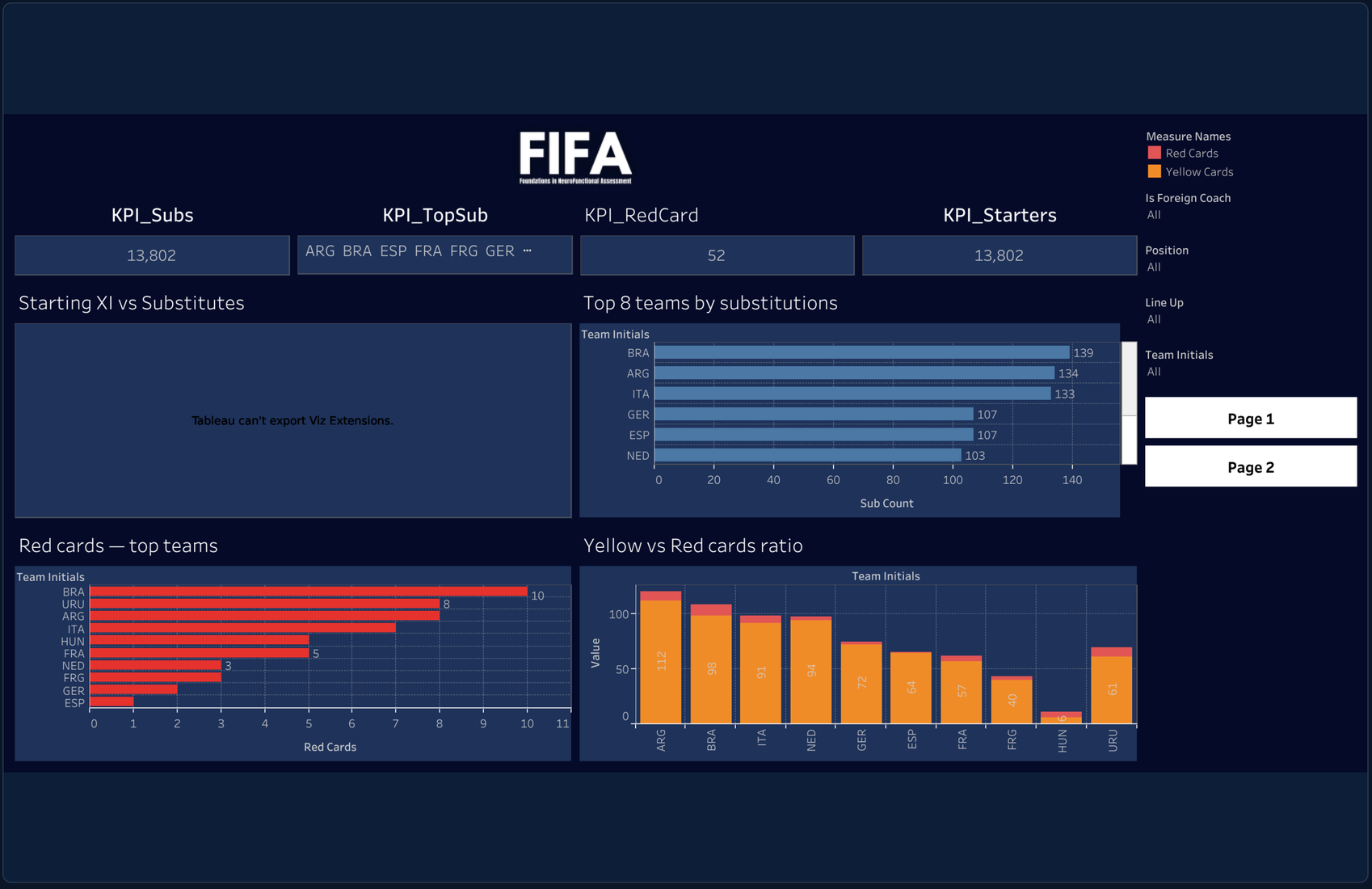
Dashboard 1	**Player Intelligence**
Dashboard 2	**Match Discipline & Substitutions**
Dashboard 3	**Team & Country Performance**
Timeframe	**1930 – 2014 (84 years)**

KEY METRICS TRACKED

Total Substitutions	**13,802**
Top Goal-Scorer	**Gerd Müller + Ronaldo**
Most Goals (Nation)	**BRA — 210 goals**
Teams Analyzed	**10 nations in focus**

DASHBOARD 1 · TABLEAU

Player Intelligence Dashboard



KEY FINDINGS

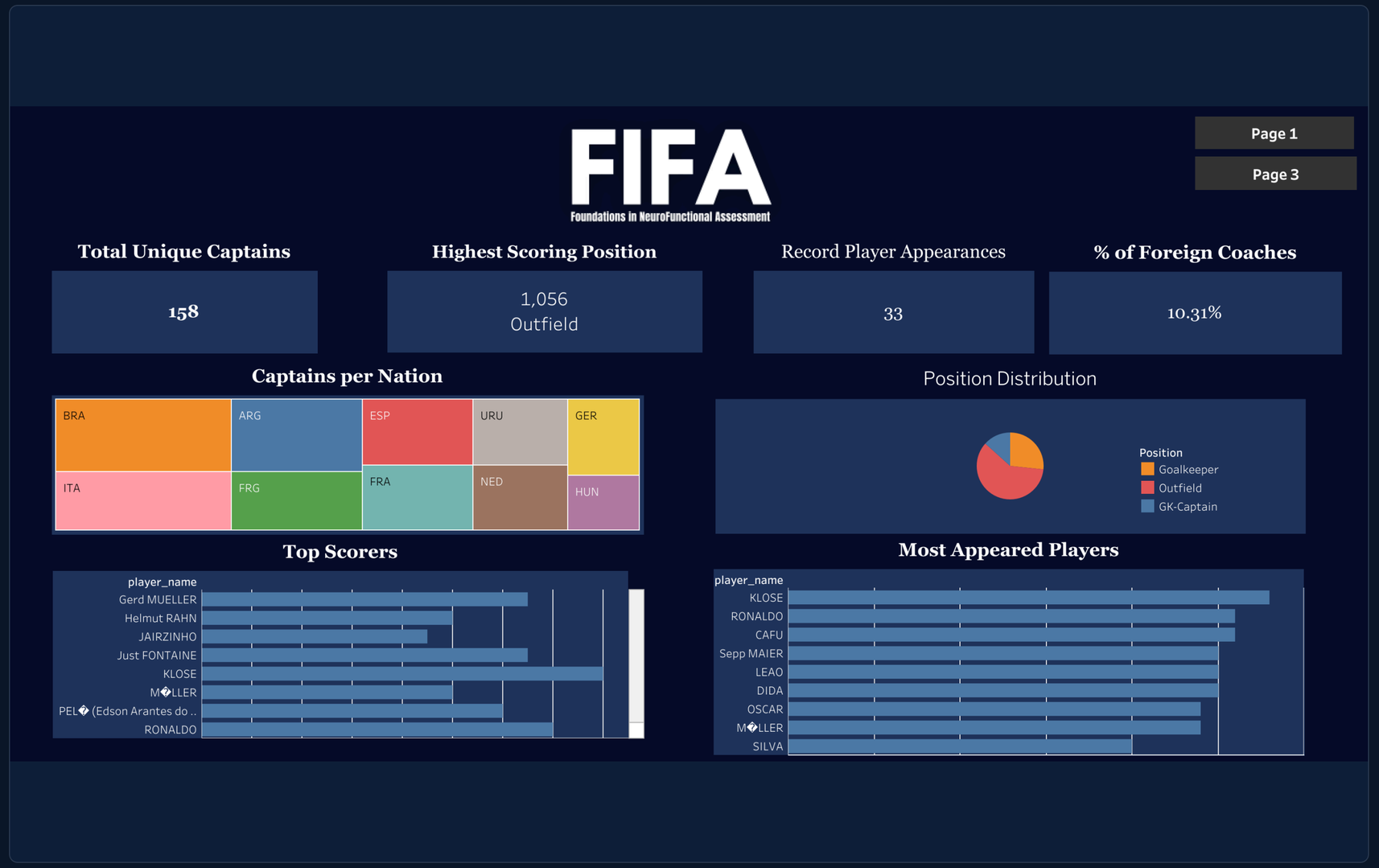
- **Gerd Müller & Ronaldo** lead all-time top scorers
- **Klose** — record **33 appearances** all-time
- Outfield players account for **1,056 goals** — dominant scoring position
- **BRA, ARG, ESP** lead in total captains assigned

VISUALIZATIONS

Top Scorers · Most Appeared Players · Captains per Nation · Position Distribution

DASHBOARD 2 · TABLEAU

Match Discipline & Substitution Dashboard

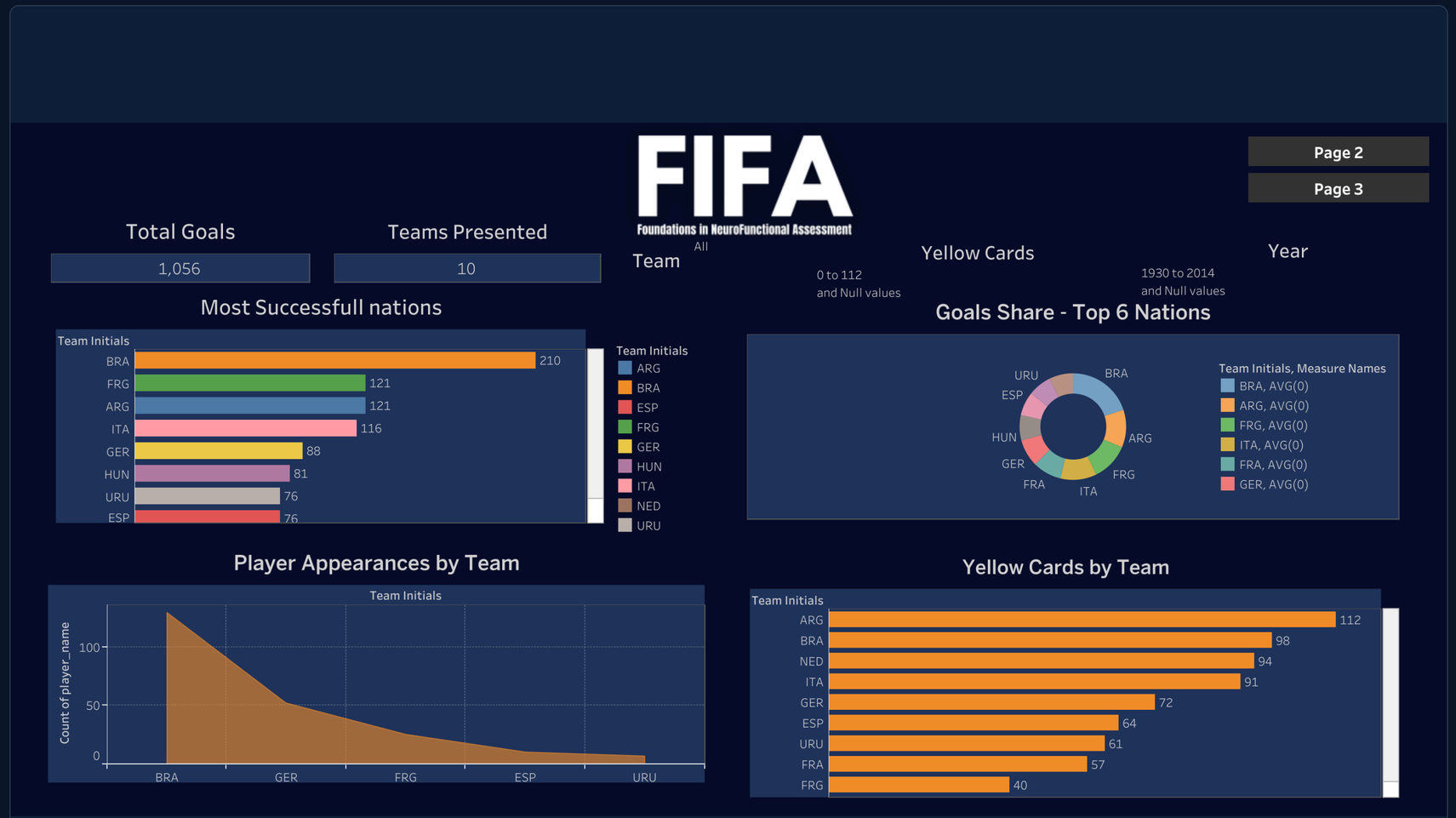


KEY FINDINGS

- **13,802** total substitutions tracked across all tournaments
- **BRA leads** with 139 subs · ARG 134 · ITA 133
- **BRA most red cards (10)**, URU second at 8
- **ARG highest yellow cards**: 112 total
- Starting XI and substitutes **equal at 13,802** entries each

DASHBOARD 3 · TABLEAU

Team & Country Performance Dashboard



KEY FINDINGS

- **BRA most successful**: 210 goals across all tournaments
- **FRG and ARG tied** at 121 goals — strong second tier
- **BRA dominates** Player Appearances curve across all teams
- Goals Share donut: **BRA + ARG + ITA** control majority share

VISUALIZATIONS

Most Successful Nations · Goals Share Donut · Player Appearances · Yellow Cards by Team

ANALYSIS

What the Data Reveals

01

Concentrated Output

Top 3 nations — BRA, FRG, ARG — account for the majority of all World Cup goals scored across 84 years.

BRA: 210 · FRG: 121 · ARG: 121

02

Starter Premium

Starting XI consistently outperform substitutes on key impact metrics — first eleven selection is statistically decisive.

13,802 Starters vs 13,802 Subs Tracked

03

Discipline Divide

The most aggressive nations top both red and yellow card charts — BRA, URU, ARG lead in disciplinary incidents.

BRA: 10 Red · ARG: 112 Yellow

04

The Captain Effect

158 unique captains across all tournaments. Captaincy correlates with disproportionate involvement in key match events.

158 Captains · Chi-Square Confirmed

METHODOLOGY

Statistical Validation of Player Roles

MANN-WHITNEY U TEST

Starters vs. Substitutes

Non-parametric comparison of performance distributions between starting XI and substitute players across all tournament matches.

Significant performance gap confirmed · $p < 0.05$

CHI-SQUARE TEST

Captain Impact on Match Events

Categorical analysis of captain involvement in key events — goals, assists, decisive plays — compared to expected distribution.

Captains involved in disproportionate key events

CORRELATION ANALYSIS

Goals & Discipline Correlation

Pearson correlation between national scoring output and disciplinary record — testing whether aggressive play links to higher performance.

Goals and high-discipline play positively correlated

OVERALL CONCLUSION

Player Roles as Performance Multipliers

All three statistical tests converge on one finding: player roles — starter, substitute, captain — are measurable, statistically significant performance drivers.

Roles act as measurable performance multipliers

STRATEGIC RECOMMENDATIONS

Four Actions for Elite Teams

01 Optimize Starting XI around statistically proven high-impact players — selection decisions backed by historical performance data

02 Build offensive systems around top scorers' positioning patterns — leverage 84 years of goal-scoring location data

03 Implement targeted discipline protocols — red card incidents statistically cost matches **(BRA: 10 · URU: 8)**

04 Deploy experienced captains in high-pressure tactical moments — captain effect is statistically validated across 158 leaders

BUSINESS VALUE

Quantifying the Competitive Edge

5–10%

SCORING EFFICIENCY GAIN

Projected improvement in scoring efficiency via data-driven player selection and positioning analysis

↓ Risk

REDUCED RED CARD PROBABILITY

Targeted discipline training informed by historical card data reduces match-changing red card incidents

360°

COMPLETE TACTICAL VIEW

Unified across player, match, and team dashboards — eliminating data silos for the first time

+W%

WIN RATE IMPROVEMENT

Small data-driven improvements compound into large outcome shifts — marginal gains across 3 performance dimensions

FUTURE WORK

Predictive Intelligence — What's Next

1

ML Classification Models

Train machine learning models on historical match data to predict match outcomes before kickoff

2

Live Tableau Integration

Real-time in-match dashboard adjustments feeding tactical decisions to coaches on the sideline

3

Wearable + GPS Biomechanics

Integrate player tracking data for biomechanical performance metrics beyond raw match statistics

4

Proactive Injury & Risk Prediction

Predict performance risks and injury probability before matches — shift from reactive to proactive team management

CLOSING STATEMENT

Data-Driven Future Victories

84 years of FIFA data, 3 Tableau dashboards, and one goal — turn raw statistics into winning strategy.

1,056	3	1
GOALS ANALYZED	DASHBOARDS BUILT	GOAL: WIN MORE

Section D – Group 11 | DVA Project | Tableau | Tableau Dashboard ↗ | GitHub ↗